Ugly Bugs

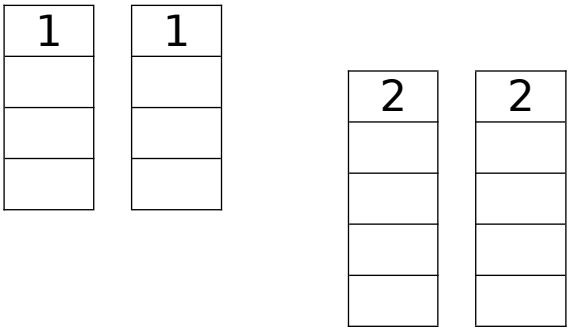
We will use an imaginary organism call the ‘ugly bug’ to see how features of an organism are inherited by its offspring.

We will be coding each ugly bug using letters for each feature. We call this code an organism’s **genotype**. We will always write our code in the same order to avoid confusion later. We will write the code into boxes which will represent **genes** on **chromosomes**.

The ugly bug has 2 pairs of chromosomes in the nucleus of each of its cells. The genes and the features they control are:

Chromosome	Dominant Allele	Gene code		Recessive Allele
1.	Round head Black eyes Long Antenna	H E A	h e a	Triangular head Red eyes Short antenna
2.	Rectangular thorax Large wings Coloured wings Hairy legs	T W Y L	t w y l	Oval thorax Small wings Plain wings Hairless legs

Fill in the **gene code** for your ugly bug, using the tables of genes above as a guide. We will assume your bug is **pure bred** or **homozygous**.

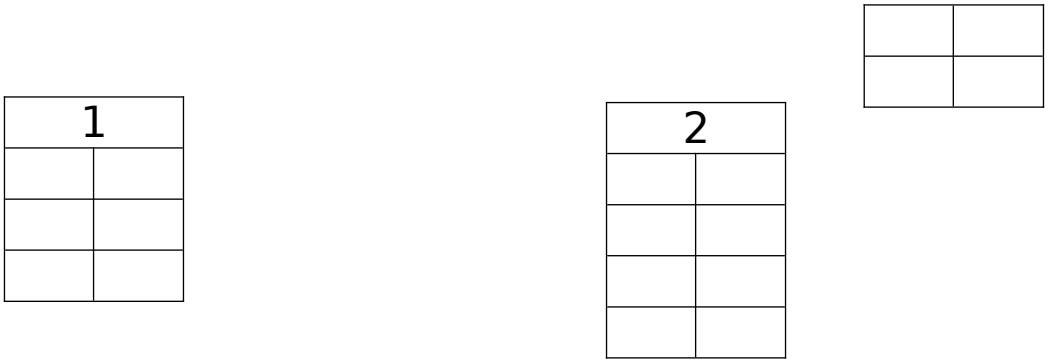


Now you bug is going to reproduce.

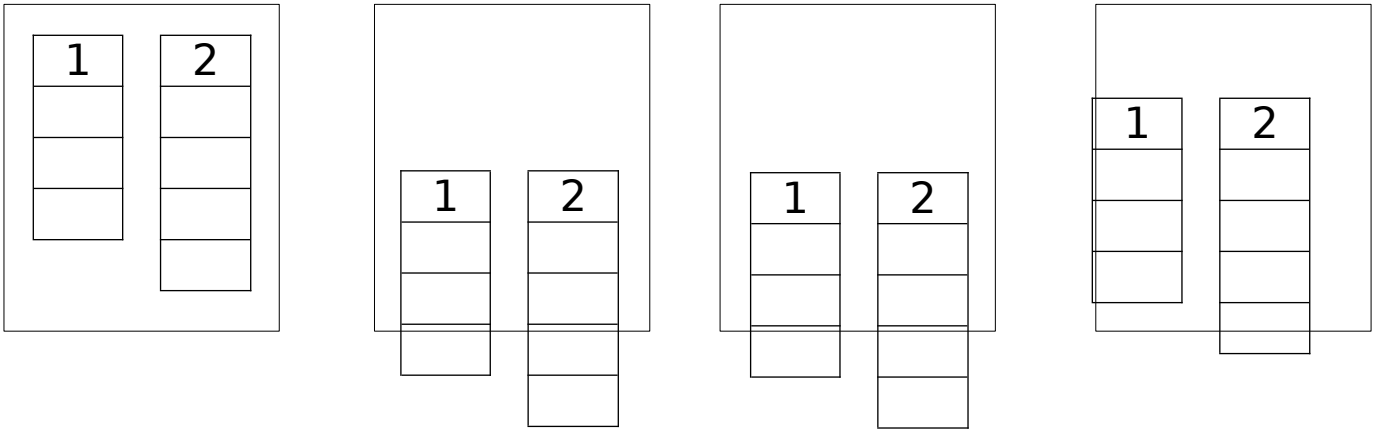
Ugly bugs reproduce sexually so they produce **gametes** by **meiosis**. Let us go through a shortened process of meiosis with the chromosomes from your ugly bug.

Step 1: The chromosomes duplicate themselves.



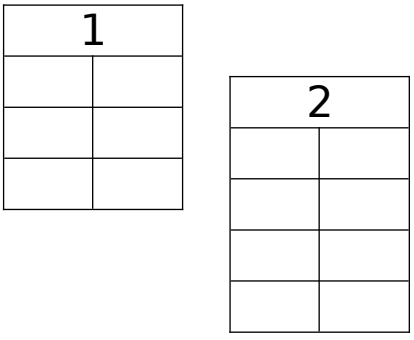


Step 2: The cell divides twice to give four gametes – each gamete contains one chromosome from each pair.



Now your ugly bug is ready to reproduce. Remember that in sexual reproduction, each parent provides a gamete. The two gametes combine to make a fertilised egg or zygote. This then divides by mitosis to grow into the new organism.

Take the gene code from one of your gametes and copy it into the first column of the chromosomes. Now copy the gene code from a gamete of a friend's ugly bug into the second column of the chromosomes.



My new ugly bug's characteristics:

You now have the code for a new ugly bug. Write the characteristics your new bug will have. Remember that both genes play a part in deciding the features of the organism. If you have two different genes for the same feature, use the table on page 1 to decide which trait is shown. Use these characteristics to assemble a picture of your new ugly bug.

In summary:
Organisms which reproduce sexually produce reproductive cells, which contain one set of _____. When they reproduce, each parent contributes a _____ to produce a fertilised egg. The _____ on each chromosome interact to determine the _____ of the new organism or offspring.

Some genes in the pair mask others and are said to be _____. The feature which is not seen when this happens is said to be _____.

When an organism has two identical genes for the same feature it is said to be a _____ or _____ feature. If it has two different genes for the same feature, it is said to be a _____ or _____.

An organism may be a pure strain in some features and a hybrid in others. You may have noticed this with your ugly bug.

Ugly bug features

Round head	or	Triangular head
Black eyes	or	Red eyes
Long antenna	or	Short antenna
Rectangular thorax	or	Oval thorax
Large wings	or	Small wings
Coloured wings	or	Plain wings
Hairy legs	or	Hairless legs